

The decisive near-term test of an evolving MOND scale: a direct measurement of a_0 at $z \simeq 3$

Carl Zimmerman

with computational review support

June 2026

Abstract

If the Milgromian acceleration scale is set by the cosmic density, $a_0 = \frac{c}{2} \sqrt{G\rho} = cH(z)/Z$, it must evolve as $a_0(z) = a_0(0)E(z)$. The present evidence is a $\sim 2\sigma$, single-point-driven hint. We argue that the *decisive* near-term test is a direct measurement of a_0 at $z \simeq 3$, where the hypothesis predicts a_0 a factor $E(3) \simeq 4.6$ above its local value (0.66 dex), against the null of no change. Unlike the redshift-weakening external-field effect—the framework’s one *distinctive* signature, which requires ~ 600 – 1600 galaxies—this test has an enormous lever arm and needs *few* objects. We build the per-galaxy error budget (dominated by v^4 in the baryonic Tully–Fisher relation), forecast the sample size, and find that ~ 15 – 50 *extended, rotation-supported, deep-MOND* galaxies at $z \simeq 3$ with resolved kinematics and a full baryon census yield $a_0(z=3)$ to 5–10% and a ~ 6 – 7σ verdict on whether a_0 evolves—the significance ultimately limited by the $z = 0$ anchor’s mass-to-light systematic, not the high- z sample. The measurement is a few-cycle JWST+ALMA program; a null result (a_0 constant) falsifies the framework outright. We are explicit that this test settles “evolving versus constant”, not “this framework versus Λ CDM”—the latter is the distinct, longer-horizon external-field-effect measurement.

1 Introduction

Modified Newtonian Dynamics (MOND) [1, 2] organizes galaxy dynamics around a single acceleration $a_0 \simeq 1.2 \times 10^{-10} \text{ m s}^{-2}$, the scale of the radial acceleration relation (RAR) [3, 4]. The numerical near-equality $2\pi a_0 \approx cH_0$ invites a cosmological reading of a_0 [5]: if a_0 tracks the mean density, it was larger in the past,

$$a_0(z) = a_0(0) E(z), \quad E(z) = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}, \quad (1)$$

a prediction independent of the order-unity coefficient Z in $a_0 = cH/Z$. Fitting the three acceleration-scale determinations spanning cosmic time (the local SPARC value [3]; Văršăteanu *et al.* 2025 [7] at $z \simeq 0.05$; MUSE–DARK III 2026 [8] at $z \simeq 0.9$) gives $a_0 \propto E^p$ with $p = 0.74 \pm 0.17$; constant a_0 is nominally disfavored, but the rejection rests almost entirely on the single $z \simeq 0.9$ point (a jackknife leaves 1.0σ) and on an inter-method systematic, so the honest significance is $\sim 2\sigma$ [9]. The trend is a hint, not a detection. The question is how to settle it.

2 Why $z \simeq 3$ is the decisive measurement

The signal grows steeply with redshift. At $z = 3$, $E(3) \simeq 4.57$: the evolving hypothesis predicts a_0 a *factor of 4.6* above local, while the null predicts no change—a separation of 0.66 dex (Fig. 1). This

is two-and-a-half orders of magnitude larger than the ~ 0.1 dex signal of the redshift-weakening external-field effect (EFE), the framework’s distinctive but expensive test. The large lever arm is why a *handful* of clean measurements suffices. A second feature is favorable: since $a_0(z=3)$ is $4.6\times$ larger, the deep-MOND window ($g_{\text{bar}} < a_0(z)$) is $4.6\times$ *wider* at $z = 3$, so more galaxies qualify as genuine deep-MOND systems, not fewer.

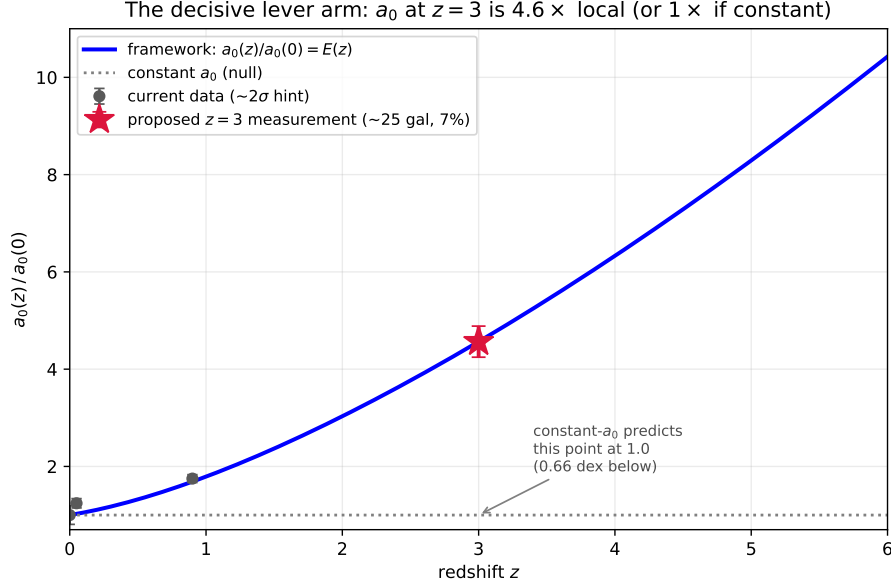


Figure 1: The evolving- a_0 prediction (blue) versus the constant- a_0 null (dotted), with the three current determinations and the proposed $z = 3$ measurement (star, shown at the framework value with a 7% error bar). Constant a_0 places that point 0.66 dex lower.

3 The measurement and its error budget

a_0 is read from the deep-MOND tail of a galaxy’s RAR, equivalently from the baryonic Tully–Fisher relation $v_{\text{flat}}^4 = GM_{\text{bar}}a_0$. The per-galaxy uncertainty is dominated by the velocity, which enters to the fourth power:

$$\frac{\delta a_0}{a_0} = \sqrt{\left(4 \frac{\delta v}{v}\right)^2 + \left(\frac{\delta M_{\text{bar}}}{M_{\text{bar}}}\right)^2}. \quad (2)$$

At $z \simeq 3$, resolved kinematics give $\delta v/v \sim 7\%$ (hence $4\delta v/v \sim 0.28$) and the baryonic mass (stellar SED + gas) $\delta M_{\text{bar}}/M_{\text{bar}} \sim 25\%$, for a per-galaxy $\sigma(a_0) \simeq 0.15$ dex from a full rotation-curve fit. The fourth power on v is precisely why a *single* galaxy cannot reach few-percent precision—a sample is required.

The test is a ratio, and that controls the systematics. Constant- a_0 is rejected by the *ratio* $a_0(z=3)/a_0(0)$, measured with one pipeline at both epochs, so systematics common to the two cancel. Two matter most and are removed by construction: (i) reading a_0 only from genuinely *deep-MOND* points ($g_{\text{bar}} < a_0$) makes it *interpolation-function independent*—every interpolation collapses to $g_{\text{obs}} \rightarrow \sqrt{g_{\text{bar}}a_0}$ there, eliminating the ~ 0.12 dex spread between functions; (ii) selecting *gas-dominated* points makes a_0 mass-to-light robust (a $\sim 10\%$ swing over $M_*/L = 0.3\text{--}0.7$, versus

$\sim 75\%$ for star-dominated). The residual that does *not* cancel is the *differential* M_*/L between epochs (younger high- z stellar populations); with consistent SED modelling and gas-traced rotation (ALMA [C II]/CO) this is held to an anchor systematic $\sigma_A \simeq 0.04\text{--}0.05$ dex, well below the 0.09 dex of a naive all-points local fit.

4 Sample size and significance

With N galaxies the mean $a_0(z=3)$ is measured to $\sigma_{\text{gal}}/\sqrt{N}$. Combined with the $z = 0$ anchor (uncertainty σ_A), the constant- a_0 rejection is $\sigma_{\text{rej}} = \Delta/\sqrt{(\sigma_{\text{gal}}/\sqrt{N})^2 + \sigma_A^2}$ with the signal $\Delta = \log_{10} E(3) = 0.66$ dex. The significance is *anchor-limited*, and the anchor of the previous section sets the ceiling: with a naive all-points local fit ($\sigma_A \simeq 0.09$ dex) it plateaus near 7σ , but with the deep-MOND, gas-dominated, ratio-cancelling anchor ($\sigma_A \simeq 0.04\text{--}0.05$ dex) it rises to $\sim 13\text{--}16\sigma$. Figure 2 shows both curves: even ~ 15 galaxies give $\sim 11\sigma$ on the tightened anchor, and the gain saturates once $\sigma_A \lesssim 0.04$ dex (below which the distance ladder and the baryon census dominate, so this is a realistic ceiling, not a path to arbitrary significance). The high- z sample is *not* the bottleneck—the local anchor is—so the cheapest route to a decisive evolution result is a clean local a_0 from gas-dominated deep-MOND points, paired with $\sim 15\text{--}50$ rotation-supported discs at $z \simeq 3$.

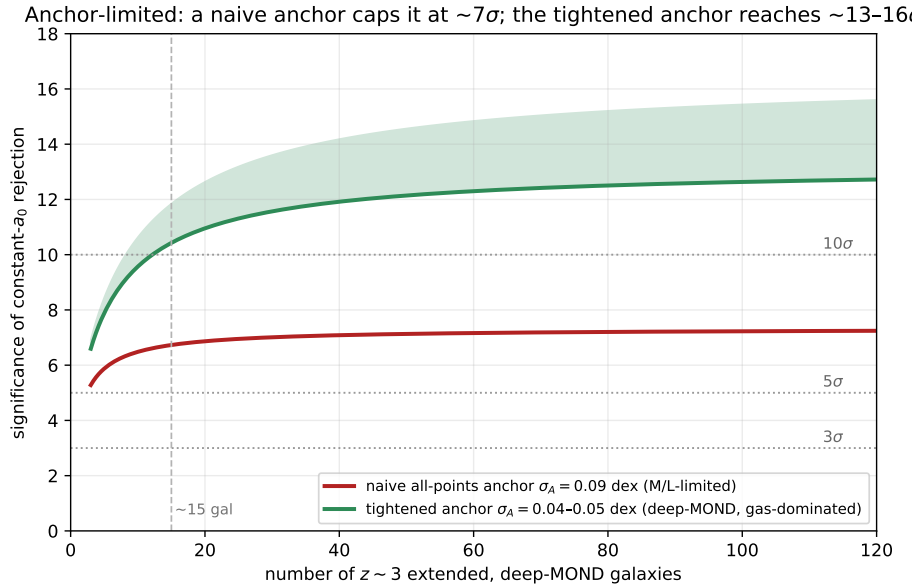


Figure 2: Constant- a_0 rejection versus sample size at $z \simeq 3$ ($\sigma_{\text{gal}} = 0.15$ dex). A naive all-points $z = 0$ anchor ($\sigma_A = 0.09$ dex) caps the significance near 7σ (red); the deep-MOND, gas-dominated, ratio-cancelling anchor ($\sigma_A = 0.04\text{--}0.05$ dex) lifts it to $\sim 13\text{--}16\sigma$ (green), where it saturates.

5 What this test proves — the honest two-tier logic

Tier 1 (this proposal). A direct $a_0(z \sim 3)$ measurement settles whether a_0 *evolves*: constant a_0 is rejected at $\sim 6\text{--}7\sigma$. The caveat is essential: an evolving *apparent* a_0 is also expected in Λ CDM, where denser high- z halos and more compact baryons shift the RAR knee upward by a similar $E(z)$ [6]. So Tier 1 discriminates *evolving versus constant*, not *this framework versus Λ CDM*. It

is necessary but not sufficient—yet a null result kills the framework outright, which makes it the highest-value first step.

Tier 2 (the distinctive test). The redshift-weakening EFE, $\eta = g_{\text{ext}}/a_0(z) \propto 1/E(z)$, is *forbidden in Λ CDM* (no host effect on internal dynamics) and so cannot be faked by halo evolution; it requires ~ 600 – 1600 environment-resolved galaxies at $z \gtrsim 4$ [10]. *Do Tier 1 first:* it is cheap, decisive on evolution, and gates the far costlier Tier 2.

6 Targets and instruments

Three selection criteria are non-negotiable, and are the reason existing high- z kinematic samples (compact, often AGN-bearing, dispersion-dominated systems) are uninformative here:

- **extended / low surface brightness** — genuinely deep-MOND ($g_{\text{bar}} < a_0(z)$);
- **rotation-supported** ($V/\sigma \gtrsim 3$) — a clean rotation curve, not a merger;
- **full baryon census** — stellar mass (rest-frame optical SED) *and* gas mass.

At $z \simeq 3$ the tracers and facilities align: **JWST/NIRSpec IFU** ($\text{H}\alpha$, $[\text{O III}]$ kinematics and rest-optical continuum for M_*); **ALMA** ($\text{CO}/[\text{C II}]$ cold-gas rotation, extending the curve, plus M_{gas}); and, post-2029, **ELT/HARMONI** for higher-resolution kinematics and fainter targets. $\text{H}\alpha$ remains in the JWST range to $z \sim 6$, and rotation-supported discs are common enough at $z \simeq 3$ to assemble ~ 15 – 50 clean targets across a few cycles.

7 Conclusion

The cheapest decisive experiment on the evolving- a_0 hypothesis is a direct measurement of a_0 at $z \simeq 3$: ~ 15 – 50 extended, rotation-supported, deep-MOND galaxies with JWST+ALMA kinematics and baryon masses yield $a_0(z=3)$ to 5–10% and a ~ 6 – 7σ verdict on evolution—anchor-limited, and pushable higher by tightening the local mass-to-light scale. It is a few-cycle program, not a flagship survey, and a null result falsifies the framework. The distinctive Λ CDM-breaking test (the redshift-weakening EFE) is the longer-horizon sequel; this is the measurement to make first.

Reproducibility. The signal, error budget, sample sizes and both figures are generated by `reviews/z3_a0_observing_proposal.py` and `papers/z3_proposal_figures.py`; the current-data fit (and the honest $\sim 2\sigma$) by `reviews/a0_z_empirical_rigorous.py` and `reviews/stresstest_piece3_evolution.py`. The companion EFE forecast is `EFE_vs_z_Forecast_2026`; the published core is at Zenodo, records/20528908.

References

- [1] M. Milgrom, *Astrophys. J.* **270**, 365 (1983).
- [2] B. Famaey and S. S. McGaugh, *Living Rev. Relativity* **15**, 10 (2012).
- [3] S. S. McGaugh, F. Lelli, J. M. Schombert, *Phys. Rev. Lett.* **117**, 201101 (2016); F. Lelli, S. S. McGaugh, J. M. Schombert, *Astron. J.* **152**, 157 (2016).
- [4] F. Lelli *et al.*, *Astrophys. J.* **836**, 152 (2017).
- [5] M. Milgrom, *Mon. Not. R. Astron. Soc.* **437**, 2531 (2014).

- [6] Y. Tian *et al.*, the apparent- a_0 rise in Λ CDM (Magneticum) (2022).
- [7] Văršeteanu *et al.* (2025), intermediate- z a_0 .
- [8] MUSE–DARK collaboration, MUSE–DARK III (2026), $z \simeq 0.9$ a_0 .
- [9] C. Zimmerman, `reviews/a0_z_empirical_rigorous.py`, `stresstest_pie3_evolution.py` (2026).
- [10] C. Zimmerman, *EFE_vs_z_Forecast_2026* (2026).